11.H1 Inscribing Polygons in Circles

Lesson Introduction

Benchmark Focus

MA.912.GR.5.4 Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons.

Learning Targets

- I can construct an equilateral triangle inscribed in a circle.
- I can construct a square inscribed in a circle.
- I can construct a regular hexagon inscribed in a circle.

Benchmark Coherence

Building On	Working Toward
MA.912.GR.5.3 Construct the inscribed and circumscribed circles of a triangle.	N/A

Essential Questions

- How do you construct an equilateral triangle inscribed in a circle?
- How do you construct a square inscribed in a circle?
- How do you construct a regular hexagon in a circle?

Lesson Narrative

In this lesson, students discover two methods for performing each of the following constructions: an equilateral triangle inscribed in a circle, a square inscribed in a circle, and a regular hexagon in a circle. Students sort cards to demonstrate their understanding of the sequence of steps needed to complete each construction using each method.

Component Summary

11.H1.1 Warm-Up (~5 min.)	Investigate solar system *(SE p. 187)*	11.H1.3 Activity (20 min.)	Sort cards with construction steps *(SE p. 189)*
11.H1.2 Activity (15-20 min.)	Make and test conjectures on how to construct a regular polygon inscribed in a circle *(SE p. 188)*	11.H1.4 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 191)*

Resources

Glossary	Materials	Additional Preparation
Key Terms: N/A Additional Terms: N/A	• Compass • Protractor • Straightedge • Larger Copies of Polygons in 11.H1.2 Activity • Cards for Card Sort • (Optional) GeoGebra	• This is an Honors lesson. Students in standard Geometry should not complete this lesson. Consider moving this activity until after the end of unit assessment if pacing is a concern.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>

Students may not understand the difference between the steps in methods 1 and 2 for each construction.
Students may not understand the difference between the steps needed to inscribe an equilateral triangle, a square, and a regular hexagon in a circle.

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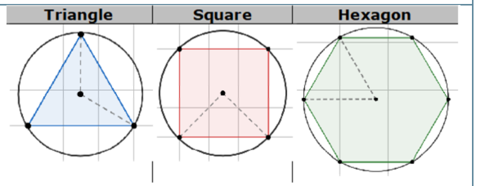
In 11.H1.2 Activity, consider modeling the constructions and then having students practice independently a few times as they develop fluency with skills.
Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the steps for each method for each construction.
Consider using a dynamic tool, such as GeoGebra, to provide a visual of each construction.

Triangle: https://www.geogebra.org/m/fbv5s3ed
Square: https://www.geogebra.org/m/Wg5aUDBk
Hexagon: https://www.geogebra.org/m/BgC68JBF

</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Think-Pair-Share
During 11.H1.2 Activity, prompt students to initially form conjectures independently. Then, ask students to discuss their conjectures with a partner. Finally, have a variety of students share their conjectures with the whole class during the debrief discussion. After each conjecture is shared, ask a variety of students if they agree or disagree and why.</td><td>MA.K12.MTR.5.1: Patterns and Structure
During 11.H1.2 and 11.H1.3 Activities, students explore different methods of constructing regular polygons inscribed in circles. Pose purposeful questions and engage students in discussions to ensure that they understand the sequences of steps for each method of each construction.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Discussions during 11.H1.2 and 11.H1.3 Activities provide an auditory entry point for students to form conjectures about how to construct each regular polygon inscribed in a circle. The diagrams in 11.H1.2 Activity provide a visual entry point for students to see each regular polygon inscribed in a circle. Provide a kinesthetic entry point by prompting students to use a compass and straightedge to check the sequences of steps they decide on in 11.H1.3 Activity by seeing if the steps result in the correct constructions.

</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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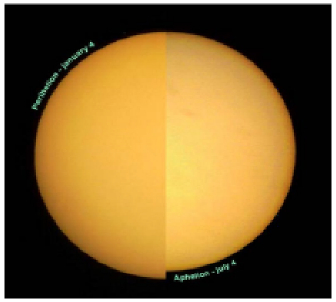
11.H1.1 Warm-Up: Earth’s Orbit

Component Overview: *Students are engaged with two interesting, real-world problems related to the diameter of the sun and the distance to the sun.*

11.H1.1 Warm-Up: Earth’s Orbit



Since Earth's orbit is an ellipse, the Earth is slightly closer to the sun in January. The picture shows two photos taken by the SOHO satellite. The left side shows the sun on January 4, 2009 and the right side shows the sun on July 4, 2009.

A scientist measured the two images. She found that the diameter of the image for January was 72 millimeters and the image for July was 69 millimeters.

1. By what percentage did the diameter of the Sun change between January and July compared to its average diameter of January and July?
   2.1%

2. If the average distance to the Sun from Earth is 149,600,000 kilometers, how much closer is Earth to the Sun in January compared to July?
   6,283,200 *kilometers*

Consider inviting an Astronomer or Physicist to join the class, virtually or in-person. Ask the guest speaker how they use geometry to help students see the utility of the topics they are studying. Also, prompt students to prepare questions for the guest speaker.

11.H1.2 Activity: Regular Polygons Inscribed in Circles

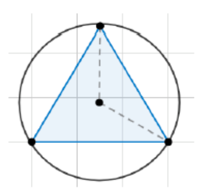
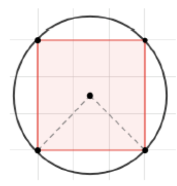
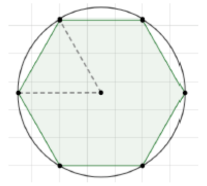
Component Overview: *Students make and test conjectures as they discover how to construct regular polygons inscribed in circles. Students will each need a copy of the polygons, compasses, and straightedges.*

11.H1.2 Activity: Regular Polygons Inscribed in Circles

Work with your partner to answer each question. Use the larger copies of each polygon provided by your teacher.

1. Use a compass to draw a circumscribed circle for each polygon.

2. Mark the center of the circle that was drawn on the polygon. Draw a central angle that passes through two subsequent vertices, as shown.

Triangle	Square	Hexagon
		

3. Use a protractor to determine the measure of each central angle.

	Triangle	Square	Hexagon
Measure of Central Angle	120°	90°	60°

Students may make errors during this activity when they are writing a conjecture about how to construct each regular polygon inscribed in a circle. Allow the opportunity for students to correct their own thinking as they work through with their partner. Do not be afraid to let the mistakes happen, only to discuss them during the transition to the next activity.

- *What do you notice about the measure of the central angle of each of the shapes?*
- *How does the measure of the central angle relate to the number of sides of the polygon?*

4. Make a conjecture about how to construct each regular polygon inscribed in a circle using a compass and a straight edge.

	Construction
Triangle	*Answers will vary.*
Square	*Answers will vary.*
Hexagon	*Answers will vary.*

5. Ask another pair of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first three columns of the table.* Have the person initial next to your summary stating that your summary was accurate.

	Other Partner Pair's Conjecture	Other Partner Pair's Initials
Triangle	*Answers will vary.*	
Square	*Answers will vary.*	
Hexagon	*Answers will vary.*	

6. Review your conjectures with your partner and if necessary, make any changes.
Answers will vary.

Think-Pair-Share
Prompt students to initially form conjectures independently. Then, ask students to discuss their conjectures with a partner. Finally, have a variety of students share their conjectures with the whole class during the debrief discussion. After each conjecture is shared, ask a variety of students if they agree or disagree and why.

11.H1.3 Activity: Card Sort

Component Overview: *Students sort cards to demonstrate their understanding of the sequence of steps needed to construct an equilateral triangle, a square, and a regular hexagon inscribed in a circle, using two different methods for each construction.*

11.H1.3 Activity: Card Sort

Sort the sets for each construction. Then, write the steps in the table.

	Construction Steps
Triangle Method 1	*1. Use a compass to draw a circle. Mark the center and label it.* *2. Keep the span of the compass used to draw the circle.* *3. Mark any point on the circumference of the circle.* *4. Place the endpoint of the compass on the marked point and make a small arc on the circumference of the circle.* *5. Continue around the circle making a total of 5 small arcs.* *6. Place a point at every other arc.* *7. Connect the points*
Square Method 1	*1. Use a compass to draw a circle. Mark the center and label it.* *2. Draw a diameter. Label the endpoints.* *3. Construct a perpendicular bisector of the diameter. Label the points that intersect with the circle.* *4. Connect the points.*

For this activity, consider allowing students to work with a partner or small group, and utilize the instructional routine "Take Turns." Each group member can take a turn at determining the next step of each construction. Students can agree or disagree with a justification and ask clarifying questions during this time.

Provide a kinesthetic entry point by prompting students to use a compass and straightedge to check the sequences of steps they decide on by seeing if the steps result in the correct constructions. Discuss the sequences of steps to provide an auditory entry point.

11.H1.3 Activity: Card Sort (continued)

Hexagon Method 1	*1. Use a compass to draw a circle. Mark the center and label it.* *2. Keep the span of the compass used to draw the circle.* *3. Mark any point on the circumference of the circle.* *4. Place the endpoint of the compass on the marked point and make a small arc on the circumference of the circle.* *5. Continue around the circle making a total of 5 small arcs.* *6. Place a point at every arc.* *7. Connect the points.*
Triangle Method 2	*1. Use a compass to draw a circle. Mark the center and label it.* *2. Keep the span of the compass used to draw the circle.* *3. Draw a diameter. Label the endpoints.* *4. Place the endpoint of the compass on one of the endpoints of the diameter.* *5. Draw a circle.* *6. Label the intersections of the new circle with the original circle.* *7. Connect the intersections with the other endpoint of the diameter.*

What are the similarities and differences between methods 1 and 2 for constructing a triangle, a square, and a hexagon inscribed in a circle?

11.H1.3 Activity: Card Sort (continued)

Square Method 2	*1. Use a compass to draw a circle. Mark the center and label it.* *2. Draw a diameter. Label the endpoints.* *3. Construct a circle with a radius larger than the original circle with center at one endpoint of the diameter.* *4. Construct a circle with a radius larger than the original circle with center at the other endpoint of the diameter.* *5. Mark the intersections of the two circles.* *6. Draw a line through the intersection of the two circles and the original circle's center.* *7. Mark the two intersections of the line and the original circle. Label the points.* *8. Connect the points.*

Consider creating, discussing, and displaying a Frayer Model to help students distinguish between the steps for each method for each construction.

Hexagon Method 2	*1. Use a compass to draw a circle. Mark the center and label it.* *2. Keep the span of the compass used to draw the circle.* *3. Draw a diameter. Label the endpoints.* *4. Place the endpoint of the compass on one of the endpoints of the diameter.* *5. Draw a circle.* *6. Label the intersections of the new circle with the original circle.* *7. Place the endpoint of the compass on the other endpoint of the diameter.* *8. Draw a circle.* *9. Label the intersections of the new circle with the original circle* *10. Connect the intersections of both circles and the endpoints of the diameter.*

Consider using a dynamic tool, such as GeoGebra, to provide a visual of each construction.

11.H1.4 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.

11.H1.4 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about inscribing polygons in circles.

   Answers will vary.